

Mechanistic Interpretability: 打开黑箱，看见 features 如何挤在一起

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

black-box model → interpretability approaches → mechanistic reverse engineering → features as directions → alignment and probes → monosemantic vs polysemantic neurons → privileged basis → superposition → toy model with sparse features → sparse autoencoders

Interpretability

- Neural networks achieve state-of-the-art results on many prediction tasks
- They have become more sophisticated and general
- At the moment, they are largely black boxes
- To increase human trust in them and avoid uninformed or risky use, we need to better understand their mechanism

Hence, the field of **interpretability**

这节课一句话

这节课的主线是：不要只看模型输入输出，要把 activation space 当作一间拥挤仓库，理解 features、neurons、basis、superposition 和 sparse autoencoder 如何解释模型内部计算。

1. 为什么要 interpretability: 黑箱越强, 风险越像雾

Interpretability

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- To increase human trust in them and avoid uninformed or risky use, we need to better understand their mechanism

Hence, the field of **interpretability**

人话直觉

把现代 LLM 想成一台很会答题、但外壳焊死的机器。它表现很好, 可我们不知道齿轮怎么咬合; 一旦它在医疗、安全、法律、自动化决策里出错, 只看最终答案就像只看体温不看病灶。

精确定义 / 机制: Slide 2 的核心是: neural networks 已经在大量 prediction tasks 上达到 state-of-the-art, 并且越来越 sophisticated/general, 但仍 largely black boxes。Interpretability 的任务就是减少这种盲用风险, 帮助我们理解、调试、信任或限制模型。考试里要把 interpretability 说成安全与可靠性的基础设施, 而不是单纯做漂亮可视化。

考试问法: 常见问法: Why is interpretability important? 答案要包含 black-box、trust、risky use、mechanism/debugging/safety。不要只写“让人类理解模型”, 要补上为什么理解会影响部署风险。

2. 四条解释路线：从看行为到拆内部电路

Approaches to interpretability

- Behavioural: model is black-box, analysing inputs and outputs
- Attributional: explain outputs by tracing predictions to individual input contributions using gradients
- Concept-based: probe learned representations for high-level concepts; also includes representation engineering
- Mechanistic: study the components of models through analysis of features, neurons, layers and connections

人话直觉

理解一个人，你可以看他考试答卷，也可以做脑成像；理解模型也一样。behavioral 像看答卷，mechanistic 像拆脑回路，中间还有 attribution 和 concept-based 两种桥。

精确定义 / 机制： Slide 4 给出四类方法：behavioral 把模型当 black box，只分析 inputs/outputs；attributional 把输出追溯到输入贡献，例如 gradients/saliency；concept-based probing learned representations 中是否有高层概念；mechanistic 则研究 features、neurons、layers、connections，目标是解释内部计算组件如何产生行为。

考试问法： 如果考 compare approaches，要按“看哪里”回答：behavioral 看 I/O，attributional 看输入贡献，concept-based 看表示是否编码概念，mechanistic 看内部组件与因果机制。

3. Mechanistic interpretability: 不是问模型会什么，而是问它怎么做到

Mechanistic interpretability

- Aims to completely specify a neural network's computation (reverse engineering)
- Focus on AI safety as a motivation

人话直觉

如果 behavioral evaluation 是试驾汽车，mechanistic interpretability 就是打开引擎盖，画出油路、电路和传动链。它想知道某个能力到底由哪些内部零件实现。

精确定义 / 机制： Slide 5 定义 mechanistic interpretability: aims to completely specify a neural network's computation，也就是 reverse engineering。它尤其以 AI safety 为动机，因为只有理解内部机制，才可能定位 deception、harmful feature、事实表示、错误 circuit，并进行监控或干预。

考试问法：考 mechanistic interpretability 定义时，要写 reverse-engineering + components/features/neurons/layers/connections + computation + safety motivation。不要把它写成普通 attention visualization。

4. Feature 是 activation space 里的方向：投影越大，味道越浓

The basic notion of features

- Features are the basic “constituent” in analysing transformer models or neural networks
- It is difficult to always identify them, but you can think of them as a “direction” that is activated given a specific type of input
- When we have a feature, we can “measure” how much an activation is aligned with it

人话直觉

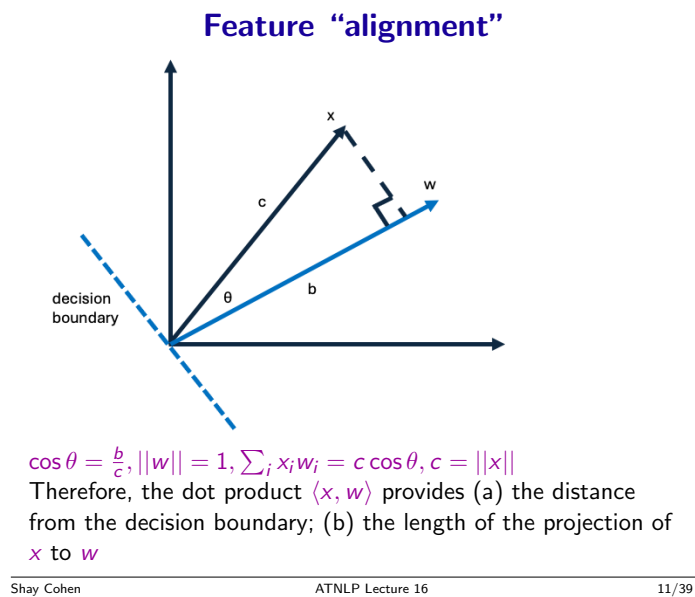
想象 activation space 是一个巨大的调味盘，某个方向代表“法语”、某个方向代表“城市名”、某个方向代表“有害请求”。输入进来后，它在某个方向上的投影越大，就越像带着那种味道。

精确定义 / 机制：Slides 6、8、9、11 建立 feature 的数学语言：vectors 是带长度的方向；dot product 衡量相似度/投影；feature 是模型内部可被特定输入激活的 direction 或 pattern。若 activation x 与 feature direction w 对齐，那么 dot product 或 projection 可作为 feature activation/alignment 的量化指标。

$$\langle x, w \rangle = \|x\| \|w\| \cos \theta, \quad \|w\| = 1 \Rightarrow \langle x, w \rangle = \|x\| \cos \theta$$

考试问法：常见问法：What is a feature? How do we measure alignment? 答：feature 是 activation space 中与某类输入/概念相关的方向；alignment 可用 projection/dot product/cosine 与 probe direction 衡量。

5. Probe 和 alignment: 能读出来, 不等于模型真的用了它



原 PPT 第 11 页

人话直觉

Probe 像拿一个探测器去 activation 里问: 这里有没有“城市名”这种信号? 探测器能读出来, 说明信息在那儿; 但不代表模型做任务时一定靠它开车。

精确定义 / 机制: Slide 10/11 的流程是: 用 activation x 预测某 feature 是否出现, 训练 logistic regression/probe; probe 的 decision boundary vector w 可当作 feature-related direction; 再用 activation 与 w 的 dot product/projection 计算 alignment。关键 caveat: probe 说明 information is linearly decodable, 不自动证明 causal use。

考试问法: 考 probe 时要写两面: 优势是检测 representation 是否包含概念; 局限是 decodability 不等于 causality。高分答案会补一句: 需要 intervention/ablation 才能证明模型真的使用该 feature。

6. Feature、concept、irreducible feature: 别把人类标签直接贴到 neuron 上

Representation types

- Features - a “direction” in the activation space, for example, corresponding to “green pixels” or “upper case letter”
- Concepts - a grouping of features or some other pattern of features that corresponds more abstractly to a concept (leaf/tree, name of a city)

人话直觉

人类说“树”是一个 concept；模型内部可能把它拆成绿色、枝干、叶片、上下文等多个 features。我们看到的是标签，但模型存的是方向组合。

精确定义 / 机制：Slide 12 说 feature 被假设为 irreducible：不能再合理分解成更基本 features 的表示原子。Slide 13 区分 feature 与 concept：feature 是 activation space 中的 direction，例如 green pixels 或 uppercase letter；concept 更抽象，可能是 features 的组合或 pattern，例如 leaf/tree/city name。

$$f \neq g(f_1, \dots, f_n) \quad \text{for simpler features } f_i$$

考试问法：如果问 feature vs concept：feature 是模型内部表示单位，concept 是人类语义层概念，二者可能相关但不等同。顺手写出 irreducible feature 是不能进一步分解的基本表示单元。

7. Monosemanticity 与 polysemanticity: 一个旋钮控制一种味道, 还是一堆味道混在一起

Monosemanticity and polysemanticity

- A neuron that fires for a single concept/feature is **monosemantic**
- It is **polysemantic** if it fires for multiple unrelated concepts/features (for example, images of cars and cats, or cities and words related to agriculture)

Food for Thought: What can we do with monosemanticity?
This is useful, for example, if we want to neutralise “bad” behaviour

人话直觉

monosemantic neuron 像一个标清楚的开关：一按就是“拒绝有害请求”。polysemantic neuron 像厨房总闸：它同时连着冰箱、灯和烤箱，关一下可能把不相关功能全影响。

精确定义 / 机制：Slides 14/15 定义：若一个 neuron 主要对应 single concept/feature，它是 monosemantic；若对多个 unrelated concepts/features 激活，它是 polysemantic。Monosemanticity 对安全有用，因为可以更直接地 neutralise bad behaviour；polysemanticity 则让解释和干预困难，因为同一 neuron 混合多个含义。

考试问法：常见题：Why is polysemanticity hard for interpretability? 答：一个 neuron 同时承载多个 unrelated features，解释不唯一，干预会产生副作用。Why monosemanticity useful? 答：更容易定位、监控、抑制单一 feature。

8. Privileged basis: 不是空间不可解释，是你坐标轴没对准

Privileged basis

An important notion with features is that of “privileged basis”

Features are correlated and intertwined, therefore we have lack of monosemanticity

A **privileged basis** is a choice of “coordinate system” that is more aligned with irreducible features (or aligns with other “semantics” the network computes), so that each axis corresponds to a meaningful feature

Neurons could become monosemantic in this case

Makes, for example, “intervention”, as mentioned, easier

人话直觉

同一张地图，歪着坐标轴看，街道全是斜线；把坐标轴转到街道方向，城市突然清楚。privileged basis 就是那套让 feature 对齐坐标轴的自然坐标系。

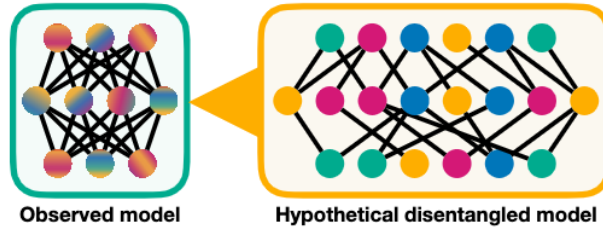
精确定义 / 机制： Slide 16 说 features correlated and intertwined，因此会 lack monosemanticity。Privileged basis 是一种 coordinate system choice，它更 aligned with irreducible features 或网络计算的 semantics，使每个 axis 对应 meaningful feature；若存在这样的 basis，neurons/coordinates 可能更 monosemantic，intervention 更容易。

考试问法：考 privileged basis 时，要强调它是 coordinate system，不是新训练一个模型。好处：features 可对应坐标轴或 neurons；如果不存在，features 会分布在任意方向，单 neuron 解释不可靠。

9. Superposition: 房间太小，模型把不常同住的 feature 塞进同一张床

Superposition

The superposition hypothesis: Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons



- Superposition: Multiple features are encoded in overlapping neuron directions, rather than each getting its own neuron (representation-level). $\Rightarrow$
- Polysemanticity: A neuron encodes multiple features / concepts at once (neuron-level).

人话直觉

如果宿舍床位少、学生多，最合理的安排不是每人一床，而是让作息不重叠的人共享床位。模型也是：feature 多、neuron/维度少，且很多 feature 不同时出现，于是它们可以叠放。

精确定义 / 机制：Slide 18 的 superposition hypothesis: neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons. Superposition 是 representation-level: 多个 features 被编码在重叠 neuron directions; polysemanticity 是 neuron-level: 一个 neuron 看起来承载多个 features/concepts。

考试问法：核心考点: superposition 和 polysemanticity 区别。答: superposition 是低维表示中 features 共享方向的编码策略; polysemanticity 是单个 neuron 表现出多个语义。前者可导致后者，但不是同一层概念。

10. Toy model: 用一个小模型解释为什么 sparsity 会鼓励 superposition

A toy model (Elhage et al., 2022)

- Investigates the notion that a network can represent more features than neurons
- Let x be a high-dimensional vector, where x_i corresponds to a feature capturing real-world concept. Assumptions about features:
 - Concept sparsity (level of sparsity S)
 - More concepts than neurons
 - Some concepts are more important than others (denoted by a_i)
- Transform x into hidden representation h and reconstruct using x' as:

$$h = Wx; x' = \text{ReLU}(W^T h + b)$$

- Evaluate using a loss function

$$\mathcal{L} = \sum_x \sum_i a_i (x_i - x'_i)^2$$

人话直觉

Toy model 像把一堆稀疏灯泡接到少量电线上：多数灯泡平时不亮，所以很多灯泡可以共享电线；只要它们不常同时亮，互相干扰就小。

精确定义 / 机制：Slide 19 建立 Elhage et al. toy model：输入 x 是高维 feature vector， x_i 对应真实概念 feature；features 有 sparsity S ，feature 数量多于 neurons，且 importance a_i 不同。模型用 W 把 feature space 压到 hidden representation h ，再用 $W^T h$ 重构 x' ，目标是按 importance 加权减少 reconstruction error。

$$h = Wx, \quad x' = \text{ReLU}(W^T h + b), \quad L = \sum_x \sum_i a_i (x_i - x'_i)^2$$

考试问法：如果问 toy model illustrates what? 答：即使简单线性重构，在容量限制、多 features、稀疏激活和 importance 加权下，superposition 会自然出现；高 importance features 更可能获得独立方向，低重要/稀疏 features 更可共享方向。

11. 为什么 increased sparsity increases superposition: 少同台, 就能共用舞台

A toy model

- Each x_i in the synthetic data is sampled based on S - the smaller the S , the more likely x_i (the feature) to be activated
- The matrix W transforms from the feature space to the “neuron” (representation) space ($rep \times feature$)
- Each row in W represents the “intensity” of the features for the corresponding row-indexed neuron. Each column provides a coverage of a feature in the neuron space.
- **Food for thought:** What does a dot-product between two rows of W indicate? How similarly two neurons respond to features. If the row is sparse, the neuron is “more monosemantic”.
- **Food for thought:** What does a dot-product between two columns of W indicate? How similarly two features are covered by neurons
- **Food for thought:** As we increase sparsity levels, will we have more superposition or less of it? Less interference among features, therefore more superposition (counter-intuitive?). If S is too high (extreme sparsity), features almost never fire, and less§ superposition - optimiser ignores dead features

Shay Cohen

ATNLP Lecture 16

19/39

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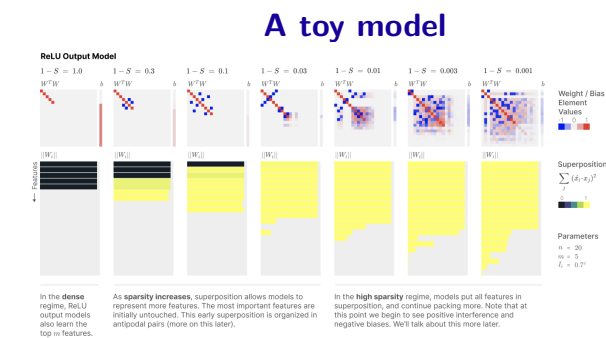
人话直觉

这就是你最开始问的那句。两个 feature 如果几乎不同时出现，它们共享同一维度也很少打架；所以越稀疏，越能把更多 feature 塞进同一表示空间。反直觉但很优雅。

精确定义 / 机制: Slides 20–25 反复追问 dot products 和 sparsity。课程里的答案是: as we increase sparsity levels, less interference among features, therefore more superposition。直观上, superposition 的成本来自同时激活时互相干扰; 若 features sparse, 不常 co-activate, 共享方向的干扰成本低。Slide 24 也提醒 extreme sparsity 例外: 如果 S 太高, features 几乎从不激活, 优化器可能忽略 dead features, superposition 反而减少或无意义。

考试问法: 考题如果问 sparsity effect: 写“higher sparsity usually allows more superposition because co-activation/interference is lower”, 再加 caveat “extreme sparsity can produce dead/ignored features”。这是高分点。

12. 读 $W^T W$ 和 column norm: 从图上看 features 有没有挤在一起



- $W^T W$ - dot products between all feature weights (to neurons). The less diagonal a matrix is, the more similarly features are distributed across neurons. We see as features become more sparse ($1 - S$ decreases), more superposition.
- $\|W_i\|$ corresponds to the norm of a column - how spread a given feature is across neurons. Yellow - very spread. The more yellow, the more features are spread (superposition)

原 PPT 第 25 页

人话直觉

$W^T W$ 像一张 feature 室友关系表: 对角线是自己, off-diagonal 是和别人挤得多不多; column norm 像某个 feature 在 neuron 空间铺了多大地毯。

精确定义 / 机制: Slide 25 解释 toy model visualization: $W^T W$ 是所有 feature weights 的 dot products。矩阵越不 diagonal, 表示 features 越相似地分布在 neurons 上, 即更多 overlap/superposition。 $\|W_i\|$ 是某个 feature column 的 norm, 表示该 feature 在 neuron space 中 spread/coverage 的程度; 图中黄色越明显, 通常表示 feature 越 spread。

$$W^T W \text{ has entries } (W^T W)_{ij} = \langle W_{:i}, W_{:j} \rangle, \quad \|W_{:i}\|_2$$

考试问法: 如果考 interpret $W^T W$: off-diagonal 项表示不同 features 在 hidden/neuron representation 中的相似或干扰; column norm 表示 feature 被保留/分布的强度。不要把 row dot product 和 column dot product 混掉。

13. Sparse autoencoders: 给模型另造一套更宽、更稀疏的可读坐标系

Cunningham et al., 2023

- Transform x into x' using feature dictionary M :

$$c = \text{ReLU}(Mx + b); \hat{x} = M^T c = \sum_i c_i f_i$$

- f_i are the rows in the “feature dictionary” M
- c gives the coefficients for how much a given feature is used. Note that M has much more rows (the autoencoder features) than columns (the dimension of x)
- The objective to train the sparse autoencoder is $\|x - \hat{x}\|^2 + \alpha \|c\|_1$
- The use of ℓ_1 norm for c is what gives the sparsity. The first term is the “reconstruction loss”

Food for thought: What are we doing here, and how is that related to “privileged basis”? We are trying to find a “privileged basis” with a larger dimension where each coordinate corresponds to some interesting coherent set of activations

Shay Cohen

ATNLP Lecture 16

22/39

原 PPT 第 28 页

人话直觉

如果原来的 neuron 坐标系太挤，SAE 就像重新画一张更大的收纳柜：格子比原维度多，但每次只打开少数格子，于是混在一起的 feature 有机会被分开贴标签。

精确定义 / 机制： Slides 26–28 介绍 sparse autoencoder: 从 residual stream/MLP/attention activations 采样，训练一个带 feature dictionary M 的 autoencoder。Encoder 产生 sparse coefficients c ，decoder 用 dictionary features f_i 加权重构 activation x 。L1 penalty 让 c 稀疏，reconstruction loss 保留信息。课程把它和 privileged basis 联系起来：我们在更高维 dictionary 中寻找一组更可解释的 feature coordinates。

$$c = \text{ReLU}(Mx + b), \quad \hat{x} = M^T c = \sum_i c_i f_i, \quad \mathcal{L} = \|x - \hat{x}\|_2^2 + \alpha \|c\|_1$$

考试问法： 考 SAE 时要写结构和 tradeoff: $c = \text{ReLU}(Mx + b)$, $\hat{x} = M^T c$; loss = reconstruction + alpha L1。它帮助解开 superposition，但 learned features 不一定真实可解释，需要验证。

考试速记版

- Interpretability 的动机：强模型是 black box，部署风险要求我们理解机制。
- 四类方法：behavioral 看输入输出；attributinal 看输入贡献；concept-based 看表示概念；mechanistic 看内部组件和连接。
- Mechanistic interpretability = reverse-engineer neural network computation，安全动机很强。
- Feature 可理解为 activation space 中的 direction/pattern；alignment 可用 dot product/projection 衡量。
- Probe 说明 concept 可被读出，不自动证明模型因果使用该 concept。

- Feature 是内部表示方向，concept 是人类语义组合；irreducible feature 是不能再合理分解的表示原子。
- Monosemantic neuron 对单一 feature 激活；polysemantic neuron 对多个不相关 features 激活。
- Privileged basis 是让坐标轴与 irreducible features 对齐的自然坐标系。
- Superposition 是 features 多于 neurons 时，模型用重叠方向编码更多 features。
- Superposition 是 representation-level；polysemanticity 是 neuron-level。
- Increased sparsity increases superposition: 越少同时激活，越少干扰，越适合共享维度；但 extreme sparsity 可能导致 dead features。
- Toy model 中 $W^T W$ 的 off-diagonal 表示 feature overlap/interference，column norm 表示 feature 在 hidden space 的保留/分布。
- Sparse autoencoder 用 reconstruction + L1 sparsity 学 feature dictionary，试图找到更高维、更可解释的 basis。

一段稳妥考场回答

Mechanistic interpretability 试图 reverse-engineer 神经网络内部计算，而不只是观察输入输出。它把 activations 看作向量空间中的表示，features 可理解为某些会被特定输入激活的 directions；feature alignment 可通过 projection/dot product 或 probe direction 估计。直接把 neuron 等同 feature 往往失败，因为有 polysemanticity：一个 neuron 可能混合多个不相关 features。Superposition hypothesis 解释了这种现象：当 features 数量多于 neurons/维度时，模型会把多个 features 编码在重叠方向中。Toy model 显示，在容量有限、features sparse 且 importance 不同的情况下，superposition 是一种高效编码策略；increased sparsity 通常 increases superposition，因为 features 更少同时激活、干扰更低，但极端稀疏可能产生 dead features。Sparse autoencoders 则通过学习更高维 feature dictionary 和稀疏 coefficients，尝试找到更接近 privileged basis 的可解释表示。